

Lindberg's L^p Jacobian Norm Equivalence Was Answered Affirmatively by Hytönen

Literature-resolution packet for arXiv:1611.02223

11 August 2026

Classification: literature already answered. Equation (7.2) of Lindberg (2017) is true for every dimension $n \geq 2$ and every $1 < p < \infty$. It follows directly from Hytönen's Theorem 3.1.4 (2021 version of arXiv:1804.11167); the specialization is also written out in Hytönen's proof of Theorem 3.2.1. This answers the norm-equivalence subproblem, not Iwaniec's stronger conjecture that every L^p datum is one Jacobian or admits a continuous choice of preimage.

1 The exact source question

In Section 7.3 of [1], after proving a first-category result for inhomogeneous Sobolev maps, Lindberg asks whether

$$\|b\|_{L^{p'}(\mathbb{R}^n)} \asymp \sup_{\|Du\|_{L^{np}(\mathbb{R}^n)} \leq 1} \left| \int_{\mathbb{R}^n} b J u \right| \quad (b \in L^{p'}(\mathbb{R}^n)) \quad (1)$$

holds for $n \geq 2$ and $1 < p < \infty$. This is equation (7.2) on PDF page 27 (published page 739). The source calls it an important step toward Conjecture 7.1, relates its planar case to an Ahlfors–Beurling commutator, and ends: “The author has been unable to prove or disprove (7.2).”

The normalization in (1) uses any fixed finite-dimensional matrix norm for Du ; changing that norm changes only the implicit constants.

2 The later theorem

Let $d \geq 2$. Hytönen's Theorem 3.1.4 [2] considers exponents $r_i \in (1, \infty)$, defined by

$$\frac{1}{r} = \sum_{i=1}^d \frac{1}{r_i},$$

and the quantity

$$\Gamma(b) = \sup \left\{ \left| \int_{\mathbb{R}^d} b J(u) \right| : u = (u_i)_{i=1}^d \in C_c^\infty(\mathbb{R}^d)^d, \|\nabla u_i\|_{L^{r_i}} \leq 1 \text{ for all } i \right\}. \quad (2)$$

For $r > 1$, the theorem says that $\Gamma(b) < \infty$ exactly when $b = a + c$ with $a \in L^{r'}$ and c constant, and that the corresponding function-space norm is comparable to $\Gamma(b)$.

Now take

$$r_1 = \cdots = r_d = pd.$$

Then $r = p$. If $b \in L^{p'}(\mathbb{R}^d)$, the additive constant must be zero, so Theorem 3.1.4 gives

$$\|b\|_{L^{p'}(\mathbb{R}^d)} \asymp \Gamma(b). \quad (3)$$

The constraint in (2) is equivalent, up to a constant depending only on d, p , to $\|Du\|_{L^{pd}} \leq 1$. Density of C_c^∞ in the homogeneous Sobolev seminorm and continuity of the Jacobian pairing identify (3) with Lindberg’s (1).

This is not merely an inferred specialization. On PDF pages 29–30, Hytönen’s proof of Theorem 3.2.1 explicitly writes the dual lower bound against normalized Jacobians and says that it is “precisely” Theorem 3.1.4 with $r = p$ and $r_1 = \cdots = r_d = pd$. Remark 3.2.3(2) then cites Lindberg’s page 739, explains that Lindberg sketched the planar case, and states that the paper proved the more general result directly with the Jacobian.

3 Verdict and scope boundary

Thus equation (7.2) has an affirmative answer in all dimensions and for the full exponent range asked in the source. Hytönen also proves in Theorem 3.2.1 that every L^p function is an absolutely controlled convergent series of Jacobians, i.e. the appropriate weak factorization.

The result does *not* show that every $f \in L^p$ is a single Jacobian. It therefore does not settle bare surjectivity of

$$J : \dot{W}^{1,pd}(\mathbb{R}^d, \mathbb{R}^d) \longrightarrow L^p(\mathbb{R}^d),$$

and it does not construct a continuous right inverse. Those are the stronger assertions in Conjecture 7.1.

Evidence locations. Source: arXiv:1611.02223, PDF page 27, equation (7.2) and the final sentence before the acknowledgements. Supporting paper: arXiv:1804.11167v4, Theorem 3.1.4 on PDF page 25; proof of Theorem 3.2.1 on pages 29–30; and Remark 3.2.3(2) on page 30.

Search and confidence. The match is explicit: the later paper cites Lindberg’s published page 739 and discusses the same Jacobian/commutator route. Classification confidence is high. The only care needed is the scope boundary between a norming/weak- factorization result and representation by one Jacobian.

References

- [1] S. Lindberg, *On the Hardy Space Theory of Compensated Compactness Quantities*, Arch. Ration. Mech. Anal. **224** (2017), 709–742; arXiv:1611.02223.
- [2] T. P. Hytönen, *The L^p -to- L^q Boundedness of Commutators with Applications to the Jacobian Operator*, J. Math. Pures Appl. **156** (2021), 351–391; arXiv:1804.11167v4.